Compiler Design
All College Assessment Questions 2026

notes.arpanadhikari7.com.np

Compiled by: Arpan Adhikari

Available on notes.arpanadhikari7.com.np

GANDAKI COLLEGE OF EINGINEERING AND SCIENCE

Level: Bachelor | Semester: Spring | Year : 2026
Programme: BE CE VI | Full Marks: 100
Course: Compiler Design | Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate between a compiler and an interpreter. What are the major phases in the structure of a compiler? 8

   b) State and explain data flow analysis equation and decide what is still useful and what is outdated in the following block segment. 7

   1: a = 5;
   2: b = a + 1;
   3: a = 7;
   4: c = a + b;

2. a) Convert the following R.E. to DFA using FOLLOWPOS method. (a+b)*abb. 8

   OR

   Convert the regular expression (a|b)*abb into an NFA using Thomson's construction, and then convert it into a DFA.

   b) Explain the terms token pattern and lexeme in compilation. What is the process of recognizing complex words using finite automata? Explain with necessary example. 7

3. a) Explain in detail about the recursive-descent parsing with suitable example. 7

   b) Given the following grammar 8

   E → TE'
   E' → +TE' | ε
   T → FT'
   T' → *FT' | ε
   F → (E) | id

   Construct the parsing table for this grammar for non-recursive predictive parser.

Available on notes.arpanadhikari7.com.np

4. a) Show that the input string 'abbcde' lies in the grammar using shift reduce parsing. 5

S → aABe
A → Abc| b
B → d

b) Compute the LR(1) collection of items from the following grammar. 10

E→BB
B→cB|d

Also construct the LR(1) parse table and use it to check if the string **ccdd$** is syntactically correct according to the above grammar.

5. a) What is three address code? Translate the expression a = b +c*d into three address code statement, quadruples, triples and indirect triples. 8

b) Explain about peephole optimization with example 7

OR

What transformation can be made on basic blocks. List and explain them.

6. a) What are the generic issues in the design of code generators? Explain. 6

b) Define Machine Independent Code optimization in Compiler Design with their types and necessary examples. 9

7. Write short notes **(Any Two)** 2×5

a) Basic blocks and Flowgraphs
b) Hash Tables and Symbol Tables
c) Define input buffering with the relevant diagram.

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

POKHARA UNIVERSITY
FACULTY OF SCIENCE & TECHNOLOGY
SCHOOL OF ENGINEERING

Exam	Final Internal Examination 2026 Spring		
Level	B.E.	F M	100
Program	Computer	PM	45
Year/ Part	3rd /VI	Time	3 Hrs

Subject: Compiler Design

Candidates are required to give answers in their own words as far as practicable.
The figure in the margin indicates full marks. Assume suitable data if necessary.
Attempt all the questions.

1. a) Explain all phases of the compiler by tracing the compilation of the statement through every phase. 7
   result = a * (b + c) - d;

   b) Explain the working of a lexical analyzer. Discuss the role of input buffering and symbol table during lexical analysis. 8

2. a) Construct a NFA using Thomson's Construction for the regular expression (a|b)*.abb and then convert it into a DFA. 8

   b) Explain how identifiers are recognized by a lexical analyzer.
   Construct a transition diagram to recognize valid identifiers, assuming that an identifier begins with an alphabet and may be followed by alphabets, digits, or underscores (_). 7

3. a) Explain the role of Context-Free Grammar (CFG) in syntax analysis. Illustrate how a parser verifies the syntactic correctness of a source program using CFG. 7

   b) Explain LL(1) parsing and the steps to construct an LL(1) parsing table for the following grammar: 8
   $E \rightarrow TE'$
   $E' \rightarrow +TE' \mid \varepsilon$
   $T \rightarrow FT'$
   $T' \rightarrow *FT' \mid \varepsilon$
   $F \rightarrow (E) \mid id$
   Parse the string ***'id+id*id'*** using the constructed parsing table.

notes.arpanadhikari7.com.np

4. a) Construct LR(1) parsing table of following grammar, 8
   S → AA
   A → aA
   A → b

   b) What is Intermediate Representation? Generate the Three Address Code (TAC) and represent it using Quadruples and Triples for the following expression. 7
   x=(a-b)*(c+d)-e/f

Available on notes.arpanadhikari7.com.np

5. a) What is a Symbol Table? Explain its use during different compilation phases. Construct the symbol table using hashing for the following declarations. 8

int total;
float average;
char grade;
double percentage;

Assume h(key)=ASCII(first character) mod 7

b) Explain the code generation process in a compiler. Discuss the challenges involved in designing an efficient code generator. 7

6. a) Explain about basic blocks mechanism. Construct the basic blocks and Control Flow Graph for the following code. 8

```
i=1
sum=0
if i>10 goto L1
sum=sum+i
i=i+1
goto 3
L1:
print(sum)
```

Identify the leaders and state whether the graph is reducible.

b) Why is code optimization important in compilation? Explain machine-independent optimization techniques. Illustrate any four optimization techniques with suitable examples. 7

7. **Write short notes on:** *(Any two)* **2×5**

a) LEX Tool
b) Register Allocation and Assignment
c) Data Flow Analysis

notes.arpanadhikari7.com.np

*** ***Best of Luck*** ***

Available on notes.arpanadhikari7.com.np

National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

Pre-University Examination

Level: Bachelor | Semester: VI Spring | Year : 2026

Program: B.E. Computer | F.M. : 100

Course: **Compiler Design** | P. M. : 45

Time : 3 Hrs.

Candidates are required to give their answer in their own words as for as practicable. The figures in the margin indicate full marks.

Attempt all questions.

1. **a)** Explain the various phases of a compiler with a suitable diagram. [7]

   b) Differentiate between a compiler and an interpreter. Explain the overview of the translation process highlighting the roles of the front end, the optimizer and the back end. [8]

2. **a)** Explain the role of a scanner (lexical analyser) in the process of compilation. How tokenization can be performed explain [7]

   Or

   Explain how a DFA is used as a recognizer. Convert a given NFA to a DFA with an example.

   b) What is Thompson's Construction? Convert the regular expression (1|0)*011 into an NFA using Thompson's Construction. [8]

3. **a)** Explain the transformation of a grammar suitable for top-down parsing with example. [7]

   Or

   Construct the LL(1) parsing table for the given grammar and parse the string bcb.

   S -> AB
   A -> aA | b
   B -> CA
   C -> cC | ε

   notes.arpanadhikari7.com.np

   b) Construct the LR(1) parsing table for the given grammar: and parse the string bdd [8]

   S -> AA
   A -> bA
   A> d

4. **a)** Explain the taxonomy of intermediate representations used in compilers. [7]

   b) What is a symbol table? Explain how a hash table is used for building a symbol table. [8]

Roshan
7 August 2026 10:46 am

Available on notes.arpanadhikari7.com.np

5. a) Discuss the issues involved in the design of a code generator. [7]
   b) Explain register allocation and assignment. Discuss the dynamic programming approach to code generation. [8]

6. a) Explain the concept of partial redundancy elimination. [7]
   b) Explain the concept of loops in flow graphs and constant propagation with an example. [8]

7. **Write short notes on: (Any TWO)** [2×5]
   a) Compiler vs Interpreter
   b) Context free grammar
   c) Peephole Optimization

notes.arpanadhikari7.com.np



Available on notes.arpanadhikari7.com.np

Nepal Engineering College
Changunarayan, Bhaktapur

Final Assessment

Level: Bachelor
Program: BE (6th Sem)
Course: Compiler Design

Year:2026
Full Mark: 100
Pass Mark: 45
Time: 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figure in the margin indicates full marks.
Attempt all the questions.

1. a. What is Compiler? Explain its phases with block diagram and example. (8)
   b. List out the tasks performed by lexical analyzer. How can compiler recognize the tokens? (7)
2. a. Convert (1|+10)* to NFA using Thompson's Construction and to DFA using subset construction method (8)
   b. How does recursive descent parsing works? Explain with example. (7)
3. a. What is parsing? Differentiate between top down parsing and bottom up parsing with examples. (7)
   b. Show that the following grammar is LL(1) or not. (8)

   S-> aBDh
   B-> cC
   C-> bC/ ε
   D->EF
   E->g/ ε
   F-> f/ ε

notes.arpanadhikari7.com.np

4. a. Define LR(1) parsing. Construct LR(1) parsing table for the given grammar: (8)

   S-> AaAb
   A-> BbBa
   A-> ε
   B-> ε

   b. What is an Intermediate Representation (IR) in compiler design, and why is it necessary? Briefly explain its types. (7)

Available on notes.arpanadhikari7.com.np

5. a. Write the simple code generator algorithm and explain it with an example. (7)

   b. What is a basic block in compiler design, and what are the rules used to identify them? Given the following three-address code, partition it into basic blocks: (8)

   1. a = b + c
   2. d = a - c
   3. if d > 0 goto L1
   4. e = d + 1
   5. goto L2
   6. L1: e = a - d
   7. L2: f = e * 2

6. a. Discuss the major issues faced in the design of a code generator in a compiler.
   Also explain the strategies used for register allocation and assignment during code generation. (8)
   b. Define data flow analysis in the context of compiler design. Explain the standard data flow equations used in this analysis. (7)

7. Write short notes on: **(Any Two)** (2x5=10)
   a. Closure properties of RE's
   b. Peephole Optimization
   c. Symbol Table

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

NEPAL COLLEGE OF INFORMATION TECHNOLOGY
Assessment Spring 2026

Level: Bachelor | Year : 2026
Program: BECE (M_D) | Full Marks: 100
Course: : Compiler Design | Pass Marks: 45
Semester:VI | Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Differentiate between Compiler and Interpreter. Explain different phases of compiler with block diagram. [10]
   b) Write about Token, Pattern, Lexeme and attributes of tokens with suitable examples. [5]
2. a) Convert b*(a + b)a to NFA using Thompson's Construction and then NFA to DFA using Subset Construction. [8]
   b) What is Lexical analysis? Explain role of Lexical Analyzer. Write about Lexical errors. [7]

   OR

   b) What is scanner? How does it recognize simple words to complex words? Explain. [7]
3. a) What is parsing? Explain different types of parsing with examples. Explain role of parser in a compiler. [7]
   b) What is LL(1) Grammar? Consider following grammar and compute FIRST and FOLLOW and also construct LL(1) parse table. [8]

   S → ACB | CbB | Ba
   A → da | BC
   B → g | ϵ
   C → h | ϵ

notes.arpanadhikari7.com.np

4. a) How do we construct LR(1) parse table? Consider following grammar and construct LR(1) parse table and parse string 'abcba' by LR(1) parsing [15]
   S → aSa
   S → bSb
   S → c

Available on notes.arpanadhikari7.com.np

5. a) Consider the statement: a = b * -c + (d-e). First find the Three Address Codes and then implement it in form of Quadruples, Triples and Indirect Triples. [8]

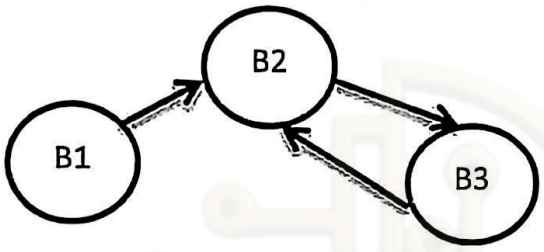
   b) Define Intermediate Representation. Explain Graphical IRs with its types. [7]

6. a) Define basic blocks and flow graph. Explain Optimization of basic blocks. [7]

OR

a) How simple code generation is done? Explain different issues in design of code generation. [7]

b) Explain steps to solve data flow equations. Consider following example and solve data flow equations. [8]



B1	1.: b=1
	2.: c=2
B2	3.: a=b+c
	4.: d=a-b
B3	5.: c=b+c
	6.: d=c-d

notes.arpanadhikari7.com.np

7. Write short notes on (*any two*): [2X5]
   a) Recursive Descent parsing
   b) Peephole Optimization
   c) Partial Redundancy Elimination

Available on notes.arpanadhikari7.com.np

COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test II

Date: 2083/04/12			
Level	BE	Full Marks	50
Programme	BEIT	Time	
Semester	VI	1.5 hrs	

Subject: - Complier Design

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *Attempt <u>All</u> questions.*
- ✓ *The figures in the margin indicate <u>Full Marks</u>.*
- ✓ *Assume suitable data if necessary.*

1. a) Discuss all the phases of the compiler with a diagram. [7]
   b) Convert given NFA to DFA subset construction. [8]

State/Input	0	1
-> q0	(q0, q2)	q0
q1	q2	q1
q2	q3	q3
*q3	ϕ	Q2

OR

Convert (a+b)*babb to NFA using Thomson's construction method.

2. a) Construct an LR(1) parsing table for the following grammar: [8]
   S' → S
   S → C C
   C → c C
   C → d
   b) What is LL(1) Grammar? Consider following grammar and compute FIRST and FOLLOW and also construct LL(1) parse table. [7]
   S → ACB | CbB | Ba A → da | BC
   B → g | ε
   C → h | ε

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

3. a) Consider the statement: a = b * -c + (d-e). First find the Three Address Codes and then implement it in form of Quadruples, Triples and Indirect Triples. [8]
   b) What is peephole optimization in a compiler? Explain common types of peephole optimizations with examples. [7]

4. Write short notes on: (*any one*) [5]
   a) Interpreter vs compiler
   b) Mapping Values to Names

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Madan Bhandari College of Engineering
Urlabari-3, Morang
Final Assessment- 2026, Spring

Level: Bachelor
Programme: BE(Computer) **Full Marks : 60**
Course: Compiler Design **Pass Marks : 30**
Year/Part:III/I **Time : 2 hrs.**

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks

1. a) Explain the complete structure of a compiler using a neat block diagram. Discuss the functions performed by every compilation phase and explain how the Symbol Table and Error Handler interact with each phase [7]
   b) Convert the regular expression (aa+bb)*a into an NFA using Thompson's Construction and then convert it into an equivalent DFA using subset construction.[8]

OR

Define Regular Expression with its operations. Convert the regular expression **(a | b)* a** into equivalent DFA by direct method.[8]

2. **a) Define Lexical Analysis.** Explain the role and function of the lexical analyzer Explain the following terms with examples: [8]
   a) Token
   b) Lexeme
   c) Pattern
   d) Lexical Error

   OR

What is the scanner? How does it recognize the simple words to complex words? Explain.

(b) Explain Input Buffering Techniques used in Lexical analysis with diagram. Discuss Single **Buffer** and **Double Buffer Schemes** with their merits and demerits.

3. **(a)** Define Top-Down Parsing.What is left Recursive grammar and how it is removed. Construct FIRST and FOLLOW sets and LL(1) parsing table for the following grammar

```
S → aBDh
B → cC
C → bC | ε
D → EF
E → g | ε
F → f | ε
```

(b) Compare Top-Down and Bottom-Up parsing with suitable examples. Construct LR(1) Parsing Table

```
S → ABC
A → aA | ε
B → bB | ε
C → a
```

4. **(a)** Define **Three Address Code (TAC), Quadruples, Triples and Indirect Triples.** Generate all representations for:

```
if (A > B)
    X = A - B;
else
    X = A + B;
```

(b) Define **Intermediate Representation (IR).** Explain **Graphical IRs** with Parse Tree and DAG. Compare **Graphical IR** and **Linear IR.**

OR

Explain **Mapping Values to Names** and describe **Symbol Table implementation using Hash Tables**

5. **a) a)** Explain **Issues in Code Generation.** Discuss **Peephole Optimization** and **Register Allocation.** **[8]**
(b) Explain Dynamic Programming Code Generation Identify Basic Blocks and draw the Flow Graph for the following three-address code. . [7]

```
1. a=b+c
2. d=a-e
3. if d>0 goto L1
4. e=d+1
5. goto L2
6. L1:e=a-d
7. L2:f=e*2
```

*****All The Best!*****

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Pokhara University

Everest Engineering College

Final Assessment

Level: Bachelor	**Year : 2026**
Program: BE CMP(6TH Sem)	**Full Marks: 100**
Section: A/B	**Pass Marks: 45**
Course: Compiler Design	**Time : 3hrs.**

- ✓ *Candidates are required to give their answer in their own words as far as practicable.*
- ✓ *The figures in the margin indicate the full marks*
- ✓ *Attempt ALL questions*

1. a) Differentiate between Compiler and Interpreter. Explain different phases of compiler with block diagram. [10]
   b) Write about Token, Pattern, Lexeme and attributes of tokens with suitable examples. [5]
2. a) Convert b*(a + b)a to NFA using Thompson's Construction and then NFA to DFA using Subset Construction. [8]
   b) What is Lexical analysis? Explain role of Lexical Analyzer. Write about Lexical errors. [7]

   OR

   b) What is scanner? How does it recognize simple words to complex words? Explain. [7]
3. a) What is parsing? Explain different types of parsing with examples. Explain role of parser in a compiler. [7]
   b) What is LL(1) Grammar? Consider following grammar and compute FIRST and FOLLOW and also construct LL(1) parse table. [8]

   $S \rightarrow ACB \mid CbB \mid Ba$

   $A \rightarrow da \mid BC$

   $B \rightarrow g \mid \epsilon$

   $C \rightarrow h \mid \epsilon$
4. a) How do we construct LR(1) parse table? Consider following grammar and construct LR(1) parse table and parse string 'abcba' by LR(1) parsing. [15]

   $S \rightarrow aSa$

   $S \rightarrow bSb$

   $S \rightarrow c$

notes.arpanadhikari7.com.np

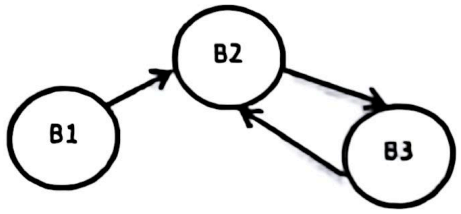
5. a) Consider the statement: a = b * -c + (d-e). First find the Three Address Codes and then implement it in form of Quadruples, Triples and Indirect Triples. [8]
   b) Define Intermediate Representation. Explain Graphical IRs with its types. [7]
6. a) Define basic blocks and flow graph. Explain Optimization of basic blocks. [7]

   OR

Available on notes.arpanadhikari7.com.np

a) How simple code generation is done? Explain different issues in design of code generation. [7]

b) Explain steps to solve data flow equations. Consider following example and solve data flow equations. [8]



B1	1.: b=1
	2.: c=2
B2	3.: a=b+c
	4.: d=a-b
B3	5.: c=b+c
	6.: d=c-d

Write short notes on *(any two)*: [2X5]

a) Recursive Descent parsing

b) Peephole Optimization

c) Partial Redundancy Elimination

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Lumbini Engineering, Management & Science College

Pre-Board Exam

2026 Spring

Level: Bachelor
Semester: 6th
Programme: BE Computer
Course: Compiler Design

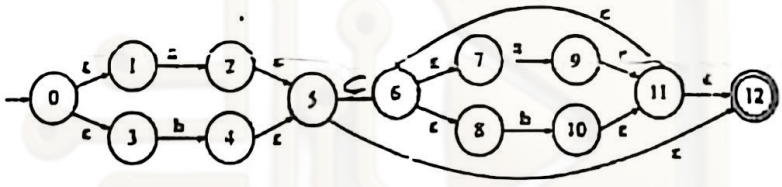
Time: 3 hrs
Full Marks: 100
Pass Marks: 45

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions

1. (a) Trace and explain the different phases of compiler with one example statement. 7
   (b) Define Regular expression. Discuss the role of lexical analyzer in detail with necessary examples. 8

2. (a) Define lexeme and token. Also produce the lexeme and corresponding tokens type with reasons to select particular token for given input. 7
   Input String: if count >= 100
   (b) Convert the given NFA to DFA 8



3. (a) What are top-down and bottom-up parsing ? Left Factoring the following grammar: 7
   S → iEiS | iEiSiS | a
   B → b
   (b) Construct the LR(1) parsing table of the following grammar. 8
   S → AaAb
   S → BbBa
   A → ε
   B → ε

notes.arpanadhikari7.com.np

4. (a) Construct parsing table for the grammar and find moves made by predictive parser on input id*(id+id) and find FIRST and FOLLOW. 7
   E→ E + T | T
   T→ T * F | F,
   F → (E) | id
   (b) First find the Three Address Code for the expression a = b * - c + (d-e) and implement it in the form of
   i. Quadruples

Available on notes.arpanadhikari7.com.np

ii. Triples

iii. Indirect Triples

5. (a) Define a Directed DAG. Construct a syntax tree and DAG: 7
a + a * (b-c) + (b-c) * d

(b) Define Leader. Convert the following program to basic block and control flow. 8
M = A + B
N = C + D
IF (M > N)
X = M – N;
ELSE
E = M + N + X

6. (a) Why code optimization is needed? Describe any three techniques for optimization. 7

(b) What is Data Flow Analysis and its application ? What are the equations used? 8

7. Write Short notes on (any two) (2 x 5 =10)

(a) Peephole Optimization

(b) Principle source of Optimization

(c) language Processing System

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

GANDAKI COLLEGE OF EINGINEERING AND SCIENCE

Level: Bachelor | Semester: Spring | Year : 2026
Programme: BE CE VI | Full Marks: 100
Course: Compiler Design | Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate between a compiler and an interpreter. What are the major phases in the structure of a compiler? 8

   b) State and explain data flow analysis equation and decide what is still useful and what is outdated in the following block segment. 7

   1: a = 5;
   2: b = a + 1;
   3: a = 7;
   4: c = a + b;

2. a) Convert the following R.E. to DFA using FOLLOWPOS method. (a+b)*abb. 8

   OR

   Convert the regular expression (a|b)*abb into an NFA using Thomson's construction, and then convert it into a DFA.

   b) Explain the terms token pattern and lexeme in compilation. What is the process of recognizing complex words using finite automata? Explain with necessary example. 7

3. a) Explain in detail about the recursive-descent parsing with suitable example. 7

   b) Given the following grammar 8

   E → TE'
   E' → +TE' | ε
   T → FT'
   T' → *FT' | ε
   F → (E) | id

   Construct the parsing table for this grammar for non-recursive predictive parser.

Available on notes.arpanadhikari7.com.np

4. a) Show that the input string 'abbcde' lies in the grammar using shift reduce parsing. 5

$S \rightarrow aABe$

$A \rightarrow Abc \mid b$.

$B \rightarrow d$

b) Compute the LR(1) collection of items from the following grammar. 10

$E \rightarrow BB$

$B \rightarrow cB \mid d$

Also construct the LR(1) parse table and use it to check if the string **ccddS** is syntactically correct according to the above grammar.

5. a) What is three address code? Translate the expression a = b +c*d into three address code statement, quadruples, triples and indirect triples. 8

b) Explain about peephole optimization with example 7

OR

What transformation can be made on basic blocks. List and explain them.

6. a) What are the generic issues in the design of code generators? Explain. 6

b) Define Machine Independent Code optimization in Compiler Design with their types and necessary examples. 9

7. Write short notes **(Any Two)** 2×5

a) Basic blocks and Flowgraphs

b) Hash Tables and Symbol Tables

c) Define input buffering with the relevant diagram.

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

POKHARA ENGINEERING COLLEGE
Final Assessment

Level: Bachelor | Semester – Spring | Year : 2026
Programme: B.E Computer | Full Marks: 100
Course: Compiler Design | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the phases of a compiler with a neat block diagram. Describe the functionality of each phase in brief. 3+5

   b) Differentiate between a compiler and an interpreter. What are the four main types of compiler? 2+5

2. a) What is the Mathematical Model of Finite Automata? Draw a non deterministic finite automate which accept 00 and 11 at the end of a string containing 0, 1 in it, e.g., 01010100 but not 000111010 3+5

   b) Explain about Thomson's Construction. 7

3. a) What are the steps for transforming a grammar for top down parsing? Explain with examples. 7

   b) Construct LL(1) parsing table for following grammar. 8

```
E --> TE'
E' --> +TE' | ε
T --> FT'
T' --> *FT' | ε
F --> id | (E)

*ε denotes epsilon
```

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

a) Describe different intermediate code forms used in compiler design. Illustrate with an example the generation of three-address code. 7

b) What is the role of symbol tables in various phases of compiler design? Generate symbol table for the following line of code: 8

```
Int x;
Float y, sum;
Sum = x+y;
```

a) What are the Issues in the design of a code generator? 7

b) Explain about register allocation in code generation? Perform register assignment for the following: 8

```
: a = 10
: b = 20
: c = a + b
: d = c + 1
: e = d + b
```

a) What is code optimization? Explain common optimization techniques like constant folding, dead code elimination, and strength reduction. 8

b) For the following grammar construct LR(1) parsing table: 7

$S \rightarrow AaAb$

$S \rightarrow BbBa$

$A \rightarrow \varepsilon$

$B \rightarrow \varepsilon$

notes.arpanadhikari7.com.np

Write short notes on **any two:** 2×5

a) Peephole optimization

b) Dynamic Programming

c) Loop Optimization

Available on notes.arpanadhikari7.com.np

Universal Engineering & Science College

Affiliated to Pokhara University

Chakupat, Lalitpur

Level: Bachelor	Semester: 6th	Year : 2026
Programme : BE Computer	Time : 3 hours	Full Marks: 100
Subject: Complier Design	**Set: B**	Pass Marks: 45

Pre-Board Examination- 2083(Spring 2026)

Candidates are required to give their answers in their own words as practicable. The figures in the margin indicate full marks.

Attempt All the questions:

1. a) What is a compiler? Explain the difference between compiler and interpreter. Discuss the role of front end, optimizer, and backend in compiler structure. [3+5]

b) Explain the phases of compiler with a suitable block diagram. Describe the function of each phase. [7]

2. a) Explain the role of lexical analyzer. Define token, lexeme, and pattern with suitable examples. [3+5]

b) Convert the following regular expression into E-NFA using Thompson's construction:
(a|b)*.a Show the construction steps. [7]

notes.arpanadhikari7.com.np

3. a) Explain parsing and its types. Remove left recursion from the following CFG: [3+5]

E → E + T | T
T → T * F | F
F → (E) | id

b) For the above grammar after removing left recursion, construct the LL(1) parsing table. [7]

Available on notes.arpanadhikari7.com.np

4. a) What do you mean by intermediate code representation? Explain Syntax Tree and DAG using the expression:

$a = b + c * d + c * d$ [2+6]

b) Write the steps for constructing a basic block. Generate three-address code for: $x = a * b + c$ [7]

5. a) What is code optimization? Explain different sources of optimization and discuss:

- Common subexpression elimination
- Dead code elimination. [1+2+5]

b) Define basic block and flow graph. Explain peephole optimization with suitable examples. [2+5]

6. a) Explain machine-independent optimization techniques. Discuss:

- Constant propagation
- Partial redundancy elimination. [2+6]

b) Define data flow analysis. Explain the importance of:

- GEN
- KILL
- IN
- OUT

in data-flow equations. [2+5]

7. Write short notes on: (Any two) [2×5]

a) Regular expression to DFA conversion

b) LR(1) parsing algorithm

c) Hash table implementation in symbol table

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

POKHARA ENGINEERING COLLEGE
Final Assessment

Level: Bachelor | Semester – Spring | Year : 2026
Programme: B.E Computer | Full Marks: 100
Course: Compiler Design | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the phases of a compiler with a neat block diagram. Describe the functionality of each phase in brief. 3+5

   b) Differentiate between a compiler and an interpreter. What are the four main types of compiler? 2+5

2. a) What is the Mathematical Model of Finite Automata? Draw a non deterministic finite automate which accept 00 and 11 at the end of a string containing 0, 1 in it, e.g., 01010100 but not 000111010 3+5

   b) Explain about Thomson's Construction. 7

3. a) What are the steps for transforming a grammar for top down parsing? Explain with examples. 7

   b) Construct LL(1) parsing table for following grammar. 8

```
E --> TE'
E' --> +TE' | ε
T --> FT'
T' --> *FT' | ε
F --> id | (E)
```

*ε denotes epsilon

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

a) Describe different intermediate code forms used in compiler design. Illustrate with an example the generation of three-address code. 7

b) What is the role of symbol tables in various phases of compiler design? Generate symbol table for the following line of code: 8

```
Int x;
Float y, sum;
Sum = x+y;
```

a) What are the Issues in the design of a code generator? 7

b) Explain about register allocation in code generation? Perform register assignment for the following: 8

```
1: a = 10
2: b = 20
3: c = a + b
4: d = c + 1
5: e = d + b
```

a) What is code optimization? Explain common optimization techniques like constant folding, dead code elimination, and strength reduction. 8

b) For the following grammar construct LR(1) parsing table: 7

$S \rightarrow AaAb$

$S \rightarrow BbBa$

$A \rightarrow \varepsilon$

$B \rightarrow \varepsilon$

notes.arpanadhikari7.com.np

Write short notes on **any two:** 2×5

a) Peephole optimization

b) Dynamic Programming

c) Loop Optimization

Available on notes.arpanadhikari7.com.np

UNITED TECHNICAL COLLEGE

Level: Bachelor | Semester: Fourth | Year : 2026
Programme: BE | Full Marks: 100
Course: Compiler Design | Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

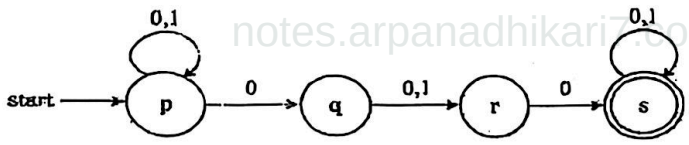
The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate between a compiler and an interpreter. What are the major phases in the structure of a compiler? 7
   b) Describe the roles of the front end, optimizer and backend in compilation. 8

2. a) Compute the FIRST and FOLLOW of all the non-terminals in the following grammar. 7
   $S \rightarrow AB$
   $A \rightarrow 0A' \mid 1A' \mid \varepsilon$
   $A' \rightarrow SSA' \mid \varepsilon$
   $B \rightarrow AS \mid 1$
   b) Construct NFA for following Regular Expressions using Thompson's Construction. 8
   i) (a + b)*a
   ii) ((a | b)cd)*

3. a) Convert NFA to DFA using subset construction 8



notes.arpanadhikari7.com.np

   b) Explain in detail about the recursive-descent parsing with example. 7

4. a) Given the following grammar 7
   $E \rightarrow TE'$
   $E' \rightarrow +TE' \mid \varepsilon$
   $T \rightarrow FT'$
   $T' \rightarrow *FT' \mid \varepsilon$

Available on notes.arpanadhikari7.com.np

F → (E) | id

Construct the parsing table for this grammar for non-recursive predictive parser.

b) Compute the LR(1) collection of items from the following grammar. 8

E → BB

B → cB | d

Also construct the LR(1) parse table and use it to check if the string ccdd$ is syntactically correct according to the above grammar.

5. a) Discuss about Directed Acyclic Graph with an example. Represent the expression A = (B + C) − (D − E) using 3AC, Quadruple and Triple. 8

b) Draw the syntax tree and DAG for the following expression: 7

(a*b) + (c-d) * (a*b) + b

6. a) What are the generic issues in the design of the code generators? Explain. 8

b) Define Machine Independent Code optimization in Compiler Design with their types and necessary examples. 7

7. Attempt any two: 2*5

a) Hash Tables and Symbol Tables

b) Context Free Grammar

c) Rules to construct the LR(1) items

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np